

TTFL: Trust-Flow-Driven Trusted Federated Learning for Edge Computing

Jiakai Zhou, Chao Bu, Shuo Zhang, Xinyu Liu
School of Computer Science and Engineering
Tianjin University of Technology, Tianjin, China
Email: wllsp0@stud.tjut.edu.cn, bc_0722@163.com

Abstract—Federated learning (FL) must assess both the execution that produced an update and the update’s effect on the global model. Under heterogeneous data, a useful long-tail client can look anomalous, whereas an attacker can build a favorable history before submitting a harmful update. Trust Flow separates hardware-anchored admission, current content evidence, historical utility, and recent risk before layer-wise aggregation. A Kalman filter handles runtime evidence; decayed Beta evidence and a risk EMA drive persistent state transitions. Layer gates, clipping, and sample-size-weighted normalization control each surviving update. The reported Flower/PyTorch experiments use CIFAR-10, ResNet-18, a replicated shared training pool, and six concurrent attack classes. Across two platforms with five runs each, the experimental record reports 92.31% clean accuracy and 10.21% macro targeted attack success rate at 30% malicious participation, with no final benign bans. A separate implementation trace provides attack-specific evaluation and gate-selection records. The evidence distinguishes final retention from participation and identifies the validation limits of the specified aggregation and calibration rules.

Index Terms—Federated learning, edge computing, trusted execution environment, risk modeling, layered aggregation, backdoor defense

1. Introduction

FL enables edge devices to train a shared model through parameter exchange while keeping their raw samples local [1], [2]. In an open edge deployment, however, the server must assess both the execution that produced an update and its effect on the global model. This assessment becomes difficult under Non-IID data: a legitimate long-tail client may repeatedly deviate from a reference direction, whereas an attacker may build a favorable record before submitting a harmful update. A single reputation score makes these cases difficult to separate. Tightening its threshold can remove useful heterogeneous contributions, while allowing past performance to compensate for current anomalies can delay the response to an attack. The resulting challenge is to preserve evidence of long-term contribution without allowing that evidence to conceal short-term risk.

Existing defenses use geometric, statistical, reference-based, or temporal evidence to flag unusual updates [3], [4],

[5], [6], [7]. Backdoor-oriented methods additionally examine model behavior, learned representations, clusters, and collusion [8], [9], [10], [11]. Such tests say little about how local training was executed, and the same tests can interpret Non-IID drift as an attack. TEE-assisted FL contributes a hardware-rooted execution boundary [12], [13], [14], [15]; admission based on that boundary still leaves the semantic safety of an admitted update unresolved.

The proposed *Trust Flow* policy addresses this tension by maintaining historical utility and short-term risk as separate states. Historical utility records contribution, while current risk can restrict an update despite a favorable earlier record. These states connect TEE-based execution admission, content auditing, and layer-wise aggregation. Their separation permits different responses to weak contribution and persistent risk, but does not by itself identify the cause of a deviation. The contributions are:

- 1) A dual-stream state representation gives historical contribution and short-term risk separate roles, supporting reversible restrictions and persistent-risk escalation.
- 2) A state-to-aggregation policy links recoverable suspicion, persistent-risk escalation, and layer-specific weighting. Content quality can lower a contribution without, by itself, forcing irreversible exclusion.
- 3) An evaluation separates final-ban outcomes, reversible review, and layer selection, combining the reported repeated-run comparison with a separately scoped implementation trace.

2. Background and Related Work

Federated learning in edge computing. Local training and server aggregation allow clients to collaborate without transferring raw samples, but leave the server dependent on the evidence accompanying each update. Heterogeneous data distributions complicate that assessment because a useful long-tail contribution may differ from the majority. Execution provenance and update quality consequently serve different purposes: the former establishes how an update was produced, while the latter determines how it should contribute to learning.

Poisoning and backdoor attacks. Sign flipping and gradient scaling disrupt optimization by changing update

directions or magnitudes. Trigger, clean-label, and semantic backdoors instead seek targeted errors while preserving plausible main-task behavior, so their updates can remain close to an expected direction [16], [17]. Across rounds, an attacker may also accumulate a favorable history before injecting a harmful update, making current behavior and past contribution potentially conflicting sources of evidence.

Active defenses and their limits. Krum and Trimmed Mean remove geometric outliers [3], [4], FLTrust weights updates against a clean direction [5], and FoolsGold and FLDetector use cross-client or cross-round evidence [6], [7]. DeepSight, FLAME, CrowdGuard, RoseAgg, FLPurifier, ShieldFL, and RaSA inspect representations, backdoor behavior, collusion, or privacy-preserving updates [8], [9], [10], [11], [18], [19], [20]. TMT-FL and related systems connect execution integrity to model training [15]. These approaches observe different evidence spaces. Trust Flow places an interface between them: attestation determines execution eligibility, privacy-aware checks expose bounded statistics, content probes inspect the current update, and the two temporal streams decide future privileges. One observation can therefore alter a layer weight, a recoverable state, or a persistent state only when the corresponding rule is satisfied. The experiments use classical methods as matched aggregation baselines and stateful variants to isolate this temporal interface.

TEE-assisted trusted training. TEEs protect execution and key material and have been used to support FL integrity and attack mitigation [12], [13], [14], [15], [21]. A quote authenticates the measured path, freshness, and update provenance. TTFL combines that hardware evidence with privacy-aware verification and cloud-side semantic auditing before assigning aggregation privileges.

3. System Model and Threat Assumptions

Figure 1 presents the client-edge-cloud environment, consisting of a central aggregation server $\mathcal{S}$ and K edge clients $\mathcal{C} = \{c_1, \dots, c_K\}$. In round t , admitted clients train from the distributed model $W^{(t-1)}$ and submit updates $\Delta W_k^{(t)}$ for auditing and aggregation. The server then computes $W^{(t)} = W^{(t-1)} + \text{Agg}(\{\Delta W_k^{(t)}\})$, with the global learning objective

$$\min_W \mathcal{L}(W) = \sum_{k=1}^K p_k \mathcal{L}_k(W), \quad p_k = |\mathcal{D}_k| / \sum_j |\mathcal{D}_j|. \quad (1)$$

The client side performs local training and supplies an attestation-bound report, while privacy-aware verification exposes the declared data and runtime statistics needed for admission and audit. Raw samples remain local. In the stated system boundary, the CPU-side TEE protects the measured admission agent, device keys, report integrity, nonce freshness, and admission state; GPU training memory and the semantic correctness of labels, samples, objectives, and updates are outside that boundary. Thus an attacker may produce a signed poisoned update inside an otherwise

measured training process, while post-attestation host tampering is excluded. Using the submitted update and this evidence, the cloud server evolves the client’s trust states and determines its aggregation privilege. Because these decisions require individual updates and audit evidence, cryptographic secure aggregation is not enabled in the evaluated protocol.

Threat model. The boundary in Fig. 1 separates report integrity from update semantics. The server is honest-but-curious and follows the aggregation protocol. It can see individual updates and audit statistics, but it does not change the stated state-transition rules. An admitted client may alter labels, samples, or optimization objectives and may conduct label flipping, trigger backdoors, clean-label poisoning, semantic backdoors, sign flipping, or gradient scaling.

The original protocol reports Intel- and AMD-capable TEE platforms. The protected path binds the identity, round, nonce, global-model digest, and received update digest before signing. The server verifies the quote/key binding, signature, freshness, and digest consistency before semantic audit. This sequence does not establish the integrity of externally collected gradient, loss, or instruction statistics. Their collection must be protected or supported by separately authenticated instrumentation. Per-platform TEE mechanisms and instrumentation are not specified in the available experiment record. The CPU-side TEE protects attestation-critical code, device keys, report integrity, nonce freshness, and the binding among client identity, round number, global-model digest, update digest, and signing key. It does not authenticate labels, samples, optimization objectives, semantic intent, or GPU memory. Consequently, the six attacks below may operate inside a valid measured training process; tampering with the update after attestation or attacks on the TEE implementation itself are excluded. Physical probing and advanced microarchitectural side channels are also excluded. The assumed malicious fraction is below 50%; the 50% experiment is a boundary stress test rather than evidence of a Byzantine guarantee. The main experiments use a 500-sample server proxy for reference and probes. The peer-reference branch only supplies a direction when that clean direction is unavailable; clean-loss and activation probes still require the proxy and are not silently treated as reference-free.

Design goals. Trust Flow aims to retain useful heterogeneous contributions while restricting persistent risk. Execution provenance, current content, historical utility, and recent risk retain separate meanings until layer-wise aggregation. Recovery and low false-ban rates are empirical goals, not consequences of attestation alone.

4. Trust-Flow Framework

Each FL round passes through four linked decision stages: admission and runtime assessment, reference-based content auditing, dual-stream state evolution, and risk-gated layered aggregation. Global model update and feedback close the five-phase loop shown in Fig. 2.

The stages in Fig. 2 pass progressively different evidence to the aggregation decision. Admission and runtime

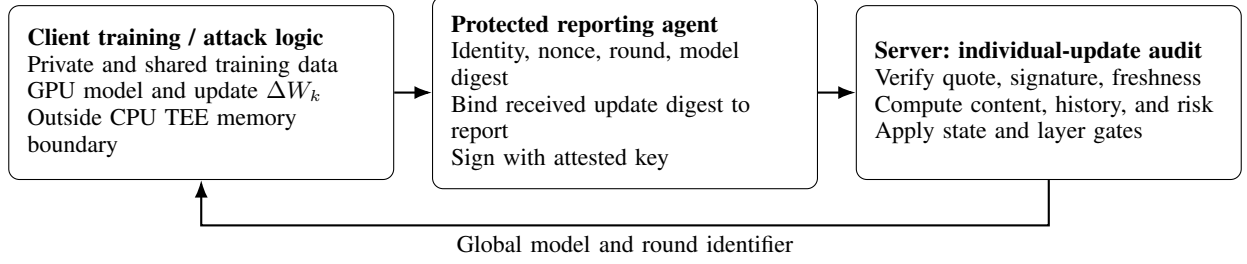


Figure 1. System and threat boundary. A signed update can contain poisoning introduced before signing. Report protection does not establish the truth of statistics collected outside the protected process. The server sees individual updates; cryptographic secure aggregation is not part of this protocol.

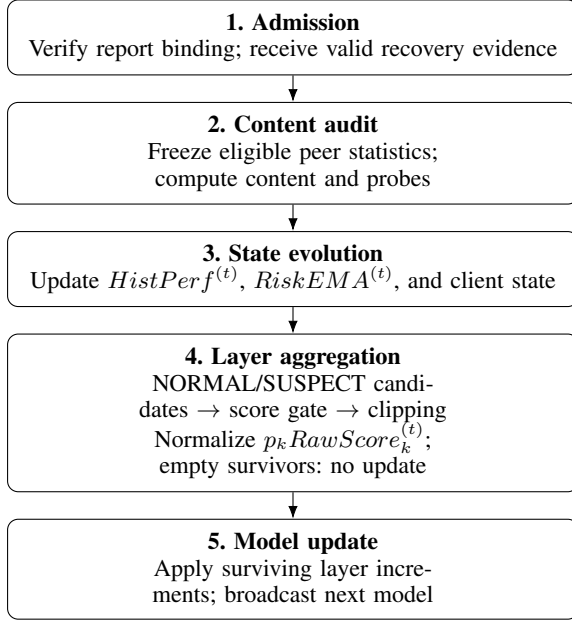


Figure 2. Round order. Quarantined submissions can supply recovery evidence but cannot enter shared peer statistics or current aggregation.

assessment produce $TrustScore$, content auditing produces $ContentScore$, and state evolution accumulates evidence in $HistPerf$ and $RiskEMA$. Their combination yields $RawScore$, which controls layer privileges. The server first removes clients that fail the admission proof, then scores the remaining updates against the reference and peer group. Keeping contribution and risk distinct allows a weak single-round contribution to reduce influence without immediately ending a benign client’s opportunity to recover.

4.1. Admission and Content Auditing

Each admitted client passes a TEE admission check. Let $M_{attest,k} \in \{0, 1\}$ indicate whether the quote, nonce freshness, code identity, round identifier, current global-model digest, update digest, and signing-key binding are valid. The runtime observation is defined from the standardized behavior sequence $\mathcal{X}_k^{(t)} = \{\Delta grad, \Delta loss, instr_dist\}$. For a declared histogram with B bins and probabilities v_b , $H = -\sum_b v_b \log v_b$ and $H_{\max} = \log B$. The entropy

is normalized as $h_k^{(t)} = H(\mathcal{X}_k^{(t)})/H_{\max} \in [0, 1]$. The observation and measurement-noise terms are

$$R_k^{(t)} = \exp(\lambda h_k^{(t)}),$$

$$Z_k^{(t)} = \text{clip}\left(w_a M_{attest,k} + (1 - w_a)(1 - h_k^{(t)}), 0, 1\right). \quad (2)$$

where $w_a = 0.7$ and $\lambda = 1.0$. For a valid proof, $Z = 1 - 0.3h$: entropy both increases measurement uncertainty and applies a soft reliability penalty to the observation mean. It is not a maliciousness label. The scalar random-walk Kalman recursion is

$$T_k^{-,(t)} = T_k^{(t-1)}, \quad P_k^{-,(t)} = P_k^{(t-1)} + Q, \quad (3)$$

$$K_k^{(t)} = \frac{P_k^{-,(t)}}{P_k^{-,(t)} + R_k^{(t)}},$$

$$T_k^{(t)} = T_k^{-,(t)} + K_k^{(t)} \left(Z_k^{(t)} - T_k^{-,(t)} \right), \quad (4)$$

$$P_k^{(t)} = (1 - K_k^{(t)}) P_k^{-,(t)}.$$

with $Q = 0.01$, $T_k^{(0)} = 1$, and $P_k^{(0)} = 0.25$. The admission score is

$$TrustScore_k^{(t)} = M_{attest,k} \text{clip}\left(\frac{T_k^{(t)}}{1 + P_k^{(t)}}, 0, 1\right). \quad (5)$$

A newly attested client starts in NORMAL; first-round evidence may reduce its current aggregation privilege but cannot directly create an irreversible blacklist.

Frozen peer statistics. Let $\mathcal{D}_t$ be valid current submissions from clients not previously blacklisted. Before updating states, define

$$\mathcal{R}_t = \{k \in \mathcal{D}_t : s_k^{(t-1)} \in \{\text{NORMAL}, \text{SUSPECT}\},$$

$$RiskEMA_k^{(t-1)} < \tau_s\}. \quad (6)$$

This revision restricts shared reference and norm statistics to $\mathcal{R}_t$. Quarantined updates are audited against those frozen statistics for recovery but do not enter them. Peer comparisons for client k exclude k itself. If required peer or proxy evidence is absent, the audit is marked incomplete: the client’s temporal states are held and its update is not aggregated. The original experiments must be checked for the same reference-set policy.

The server next constructs a reference direction. In the reported protocol, $g_{\text{root}} = g_{\text{root_clean}}$ is obtained from a

server-held balanced proxy set of 500 CIFAR-10 test samples (50 per class), separate from client training. The direction must have the sign of a model increment, for example $-\eta_{ref} \nabla_W \mathcal{L}_{proxy}(W)$ with $\eta_{ref} > 0$, because updates are applied as $W + \Delta W$. The exact reference optimizer and the proxy/evaluation split identifiers remain run-configuration requirements. If a clean direction is unavailable, the fallback is the normalized weighted mean of updates in $\mathcal{R}_t$, with $\omega_k^{ref} = TrustScore_k(1 - RiskEMA_k^{prev})^2$. This fallback changes only the reference direction; the clean-loss and activation probes still require proxy inputs. For each update, content quality combines reference and peer consistency:

$$S_{contrib,k} = \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, g_{root})), \quad (7)$$

$$\rho_{kj} = \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, \Delta W_j)), \quad (8)$$

$$S_{consist,k} = \frac{\sum_{j \in \mathcal{R}_t \setminus \{k\}} TrustScore_j \rho_{kj}}{\sum_{j \in \mathcal{R}_t \setminus \{k\}} TrustScore_j + \varepsilon}, \quad (9)$$

$$ContentScore_k = \frac{(1 + \beta_{fusion}^2) S_{consist,k} S_{contrib,k}}{D_k}, \quad (10)$$

$$D_k = \beta_{fusion}^2 S_{consist,k} + S_{contrib,k} + \varepsilon. \quad (11)$$

Here $\alpha_1 = 0.6$, $\alpha_2 = 0.4$, and $\beta_{fusion} = 2.0$. The score describes current update quality; it is not itself a permanent reputation decision.

4.2. Dual-Stream State Evolution

Cold start and first-round evidence. At $t = 1$, $HistPerf_k^{(0)}$ and $RiskEMA_k^{(0)}$ represent initial conditions rather than accumulated client behavior; the runtime trust estimate and covariance likewise start from their filter initialization. In the reference-assisted experimental setting, the server can construct g_{root_clean} without a client history. Once the first local updates arrive, runtime reports, reference alignment, peer consistency, and the available risk probes provide current-round evidence. These checks do not require a previously established reputation: they assess the first submitted update, while the two temporal streams begin accumulating evidence for later decisions.

A low content score can reflect either a useful client's skewed data or a harmful update. Since those cases require different participation decisions, Trust Flow retains historical utility and short-term risk separately. Let $q_k^{(t)} \in [0, 1]$ denote normalized content quality. With the coefficients in Table 2 and nonnegative trust weights, $ContentScore_k^{(t)}$ already lies in this interval and serves directly as $q_k^{(t)}$. The unit-evidence form of the decayed Beta model adds $q_k^{(t)}$ to positive evidence and $1 - q_k^{(t)}$ to negative evidence:

$$\begin{aligned} \alpha_k^{(t)} &= \beta_h \alpha_k^{(t-1)} + q_k^{(t)}, \\ \beta_k^{(t)} &= \beta_h \beta_k^{(t-1)} + 1 - q_k^{(t)}. \end{aligned} \quad (12)$$

Here $0 < \beta_h < 1$ is the history retention factor, distinct from the negative-evidence mass, and the reported protocol

initializes $\alpha_k^{(0)} = \beta_k^{(0)} = 1$, so $HistPerf_k^{(0)} = 0.5$. The historical utility is

$$HistPerf_k^{(t)} = \frac{\alpha_k^{(t)}}{\alpha_k^{(t)} + \beta_k^{(t)} + \varepsilon}. \quad (13)$$

Decay allows later observations to revise the contribution history rather than preserving early evidence indefinitely.

After its initial transient, the history channel retains evidence across rounds. For a legitimate long-tail client, low similarity can lower aggregation privilege while leaving later rounds available for recovery. The risk channel instead retains recent warnings through $RiskEMA_k^{(t)} = \beta_r RiskEMA_k^{(t-1)} + (1 - \beta_r) Risk_k^{inst}$, where $Risk_k^{inst}$ is the maximum current probe. Taking the maximum preserves a strong warning across probe channels; maintaining risk separately prevents a favorable history from altering that risk estimate. The state-transition rules then determine whether the evidence warrants review, quarantine, or permanent exclusion.

The separation changes how borderline clients are treated. A client with weak content quality but no rising risk can remain in the system with reduced influence, giving later rounds an opportunity to clarify whether the deviation was benign. A client with favorable historical utility but rising attack risk instead faces restricted aggregation privileges. Low historical utility can lead to temporary soft isolation, while the risk stream records sustained anomalies separately from the contribution deficit.

The two streams meet only when computing aggregation privilege:

$$\begin{aligned} RawScore_k^{(t)} &= (TrustScore_k^{(t)})^\alpha \\ &\quad \times (ContentScore_k^{(t)})^\beta (HistPerf_k^{(t)})^\gamma \\ &\quad \times (1 - RiskEMA_k^{(t)})^p. \end{aligned} \quad (14)$$

The resulting state machine distinguishes NORMAL, SUSPECT, QUARANTINE, and BLACKLIST (Table 1). Historical utility does not cancel the risk estimate. However, geometric deviation affects both content quality and risk; separate states therefore do not imply independent evidence or guaranteed discrimination between heterogeneity and attacks.

The current-round states enter the reported aggregation decision. Evidence is computed first, the state transition is applied second, the candidate set is formed third, and layer filtering and aggregation are performed last. A newly attested client can therefore lose current-round influence without being permanently banned; BLACKLIST becomes a gate on subsequent rounds.

Let $u_k^{(t)} = 1 - HistPerf_k^{(t)}$ and $r_k^{(t)} = RiskEMA_k^{(t)}$. The thresholds $(\tau_h, \tau_s, \tau_q, \tau_b) = (0.26, 0.64, 0.80, 0.90)$ refer to contribution deficit, persistent review risk, instantaneous quarantine risk, and persistent blacklist risk, respectively.

The instantaneous quarantine branch applies to clients entering the round in SUSPECT, as in the original state

TABLE 1. STATE-TRANSITION ORDER USED IN EVERY COMMUNICATION ROUND.

From	Trigger and persistence	Action
NORMAL	$u > \tau_h$ or $r \geq \tau_s$ for 2 submitted rounds	SUSPECT; current update remains conditional
SUSPECT	same condition for 3 rounds, or $Risk^{inst} \geq \tau_q$	QUARANTINE; current update excluded
QUARANTINE	$r \geq \tau_b$ for 4 rounds	BLACKLIST; next-round gate
SUSPECT or QUARANTINE	$u \leq \tau_h$ and $r < \tau_s$ for 2 rounds	recover one state level

rule. For NORMAL clients, instantaneous risk still affects the current risk EMA and score. To make the transition order explicit, this revision specifies at most one transition per submitted round, with escalation taking precedence over recovery. Recovery is QUARANTINE $\rightarrow$ SUSPECT or SUSPECT $\rightarrow$ NORMAL. Counters are local to the starting state, reset on a state change or a failed condition, and advance only on valid submissions. Missing submissions freeze historical evidence, risk, and counters. BLACKLIST excludes the current update and rejects future submissions; it is irreversible within a run. These timing conventions require checking against the implementation that produced the retained results.

4.3. Risk-Gated Layered Aggregation

Backdoor effects may concentrate in deeper layers even when shallow features retain useful heterogeneous information. Layer sensitivity sets both the score gate and clipping radius. With L layers and $l = 1, \dots, L$, the original sensitivity components are

$$S_{priv}^{(l)} = \exp(-\tau_p l / L), \quad (15)$$

$$S_{util}^{(l)} = \frac{\|g_{root}^{(l)}\|_2}{\max_m \|g_{root}^{(m)}\|_2 + \varepsilon}, \quad (16)$$

$$S_{sec}^{(l)} = \frac{1}{2} \left(1 - \frac{1}{|\mathcal{T}_l|} \sum_{k \in \mathcal{T}_l} \cos(\Delta W_k^{(l)}, g_{root}^{(l)}) \right). \quad (17)$$

Here $\mathcal{T}_l$ contains aggregation candidates with previous risk below τ_s ; if empty, the original rule uses the aggregation candidates for these statistics. If neither set contains an update, the layer remains unchanged. C_{base} is the median update norm over the same nonempty set. A zero median is handled by leaving the layer unchanged. The indexing and numerical stabilizers, including ε_c , must be recorded in the run configuration.

The privacy term decreases with depth, the utility term reflects the relative reference-gradient norm of the layer, and the security term measures deviation from the reference direction. The combined sensitivity is

$$S_{total}^{(l)} = w_p S_{priv}^{(l)} + w_u (1 - S_{util}^{(l)}) + w_s S_{sec}^{(l)}, \quad (18)$$

where w_p, w_u, w_s are the nonnegative component weights. The resulting sensitivity determines both the admission threshold and clipping radius:

$$\theta^{(l)} = \mu_{base} + \lambda_s S_{total}^{(l)}, \quad C^{(l)} = \frac{C_{base}}{S_{total}^{(l)} + \varepsilon_c}. \quad (19)$$

A layer showing stronger suspicious drift therefore has a higher entry threshold and a smaller clipping radius. Here $C_{base} > 0$ is the base update-norm scale, while $\varepsilon_c > 0$ stabilizes the denominator. Layers that remain close to the reference retain more capacity for benign heterogeneity.

Audit eligibility and aggregation eligibility. A quarantined client may supply recovery evidence without entering the frozen peer reference. Let $\mathcal{E}_t \subseteq \mathcal{D}_t$ contain submissions with a complete audit and $s_k^{(t)}$ denote the resulting state. The aggregation candidates are

$$\mathcal{A}_t = \{k \in \mathcal{E}_t \mid s_k^{(t)} \in \{\text{NORMAL}, \text{SUSPECT}\}\}. \quad (20)$$

SUSPECT retains conditional eligibility under the score and layer gates, whereas QUARANTINE and BLACKLIST have no aggregation weight. State transitions are applied before forming $\mathcal{A}_t$: a quarantine decision removes the current update, and the current-round score is used for the remaining candidates. Recovery restores eligibility only after the state transition has occurred.

The round therefore proceeds from valid submissions to current evidence, state updates, candidate selection, and finally layer filtering and aggregation. The surviving clients are $\Phi_t^{(l)} = \{k \in \mathcal{A}_t \mid \text{RawScore}_k^{(t)} \geq \theta^{(l)}\}$, written as $\Phi^{(l)}$ below for brevity. For a nonempty survivor set, their weights are re-normalized:

$$\begin{aligned} \tilde{w}_k^{(l)} &= \frac{p_k \text{RawScore}_k^{(t)}}{\sum_{j \in \Phi^{(l)}} p_j \text{RawScore}_j^{(t)} + \varepsilon}, \\ \widehat{\Delta W}_k^{(l)} &= \frac{\Delta W_k^{(l)}}{\max(1, \|\Delta W_k^{(l)}\|_2 / C^{(l)})}. \end{aligned} \quad (21)$$

The final layer update is $\Delta W_{global}^{(l)} = \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \widehat{\Delta W}_k^{(l)}$. When $\Phi^{(l)}$ is empty, the layer is left unchanged; no client is backfilled across the layer gate. This rule prevents an empty gate from being bypassed by selecting the largest raw score, and a quarantined or blacklisted client therefore never contributes through a fallback. The data-size factor p_k retains sample-size information in the credibility-adjusted aggregation, while filtering and clipping mean that the original unfiltered objective is not claimed to be preserved exactly. For survivor mass $Z_l = \sum_{j \in \Phi^{(l)}} p_j \text{RawScore}_j$, the normalized weights sum to $Z_l / (Z_l + \varepsilon)$. This compensates for removed mass up to the denominator stabilizer; it is score normalization, not a simplex projection.

Layer-wise control allows the same client update to receive different treatment across the network. When a semantic backdoor mainly alters deep features, sensitivity-dependent thresholds and clipping can constrain those layers while allowing more of a consistent shallow-layer update

to contribute. Re-normalization addresses a separate optimization effect: removing clients also removes aggregation weight, which would otherwise reduce the update magnitude merely because filtering occurred. The normalized survivor weights separate that magnitude effect from the intended clipping operation.

Cold-start gate bound.. With the stated initialization and the most favorable first-round evidence, the equations imply

$$\begin{aligned} P^{(1)} &= 0.26/1.26, \quad \text{HistPerf}^{(1)} \leq 1.9/2.8, \\ \text{RawScore}^{(1)} &\leq (1 + 0.26/1.26)^{-3} \sqrt{1.9/2.8} \\ &\approx 0.46922. \end{aligned} \quad (22)$$

Thus $S_{total}^{(l)} > 0.05769$ rules out first-round survivors under $\theta^{(l)} = 0.4 + 1.2S_{total}^{(l)}$. These are analytical consequences of the reported parameters, not measured gate frequencies. The threshold range can also exceed one, above every possible raw score. Removing the original highest-score fallback therefore changes an operationally significant branch; per-layer survival and skipped-update counts are needed to validate it.

Interpretation of the history stream.. The decayed Beta evidence has a finite effective mass and therefore behaves as a startup-corrected exponential smoother: early observations have larger gain, while later evidence is retained with factor β_h . Its first-observation gain is $1/2.8$, whereas the risk EMA gain is 0.15 ; the history stream is slower only in the steady-state sense (gain approaching 0.1). Bounded observations keep states bounded; recovery additionally requires subsequent evidence to satisfy the recovery conditions. Neither boundedness nor separate state storage guarantees recovery.

4.4. Risk Probe Computation

All instantaneous risk channels are mapped to $[0, 1]$ before fusion. For a candidate update, the geometric and amplitude probes are

$$r_{\text{grad},k}^{(t)} = \text{clip}\left(\frac{1 - \cos(\Delta W_k, g_{\text{root}})}{2}, 0, 1\right), \quad (23)$$

$$r_{\text{amp},k}^{(t)} = \text{clip}\left(\frac{\|\Delta W_k\|_2}{\text{median}_{j \in \mathcal{R}_t} \|\Delta W_j\|_2 + \varepsilon} - 1, 0, 1\right). \quad (24)$$

The clean-loss probe uses the server proxy:

$$r_{\text{probe},k}^{(t)} = \text{clip}\left(\frac{\mathcal{L}_{\text{proxy}}(W + \Delta W_k) - \mathcal{L}_{\text{proxy}}(W)}{\delta_{\text{clean}}}, 0, 1\right), \quad (25)$$

where $z_j = \max\{0, \mathcal{L}_{\text{proxy}}(W + \Delta W_j) - \mathcal{L}_{\text{proxy}}(W)\}$ is computed on trusted calibration updates and $\delta_{\text{clean}} = \max\{\varepsilon_\delta, \text{median}_j(z_j) + 3\text{MAD}_j(z_j)\} > 0$, with $\varepsilon_\delta > 0$ and $\text{MAD}(z) = \text{median}|z - \text{median}(z)|$. The positive-part calibration is a revised specification; its calibration updates and scale must be supplied before attributing measured

results to it. The activation-drift probe follows the original softmax-normalized activation definition, with $\mathcal{H}(A) = \text{softmax}(\text{vec } A)$ on the same proxy inputs and the same monitored representation:

$$r_{\text{trigger},k}^{(t)} = \text{clip}\left(\frac{\text{KL}(\mathcal{H}(A_{\text{base}}) \parallel \mathcal{H}(A_k))}{\delta_{\text{trigger}} + \varepsilon}, 0, 1\right). \quad (26)$$

Here $\delta_{\text{trigger}} > 0$ is the original trusted activation-drift scale. The monitored layer, calibration updates, and numeric scale are unresolved configuration details; a 32-bin histogram is not assumed. The probe does not require knowledge of the attacker's trigger. Peer disagreement uses the original weighted mean of pairwise cosines:

$$\begin{aligned} \bar{c}_k &= \frac{\sum_{j \in \mathcal{R}_t \setminus \{k\}} \text{TrustScore}_j \cos(\Delta W_k, \Delta W_j)}{\sum_{j \in \mathcal{R}_t \setminus \{k\}} \text{TrustScore}_j + \varepsilon}, \\ r_{\text{peer},k} &= \text{clip}(1 - \bar{c}_k, 0, 1). \end{aligned} \quad (27)$$

Cosines with a zero-norm argument are assigned zero in the revised definition. The instantaneous risk is

$$\begin{aligned} \text{Risk}_k^{\text{inst}} &= \max\{r_{\text{grad},k}, r_{\text{amp},k}, r_{\text{probe},k}, \\ &\quad r_{\text{trigger},k}, r_{\text{peer},k}\}, \\ \text{RiskEMA}_k^{(t)} &= \beta_r \text{RiskEMA}_k^{(t-1)} + (1 - \beta_r) \text{Risk}_k^{\text{inst}}. \end{aligned} \quad (28)$$

with $\beta_r = 0.85$. The maximum preserves a strong warning while the common range bounds each channel but does not equalize false-alarm probabilities. At zero cosine, the direction probe is 0.5 while the peer probe can be 1 . Per-channel distributions for benign IID, benign heterogeneous, and malicious updates are therefore needed to assess the maximum fusion rule.

4.5. Assurance Composition and Reference Role

The evidence sources complement one another because each resolves a different uncertainty. Attestation and privacy-aware verification establish execution provenance and bounded statistics for admission, while cloud auditing examines the submitted update against reference and peer evidence. Temporal states retain the distinction between a transient deviation and persistent risk. Passing the measured execution path therefore leads into semantic auditing; it does not bypass it. A single low-similarity observation can reduce contribution without causing permanent exclusion, whereas sustained direction or probe risk can still drive quarantine and later blacklisting through the state machine.

A clean validation gradient supplies a stable reference when available, while the peer-reference branch constructs a guiding direction from admitted updates when it is unavailable. Reference agreement contributes to content quality alongside peer consistency; historical contribution and recent risk then influence participation. The reference is therefore one input to auditing rather than a substitute for the dual-stream policy.

4.6. From Separate States to Participation Decisions

The distinction between historical contribution and recent risk matters most for clients near a participation boundary. A client with low utility but low risk may receive reduced influence or reversible review, allowing later evidence to clarify whether its deviation was benign. A client with substantial past utility and rising risk instead faces restrictions despite that history. Persistent high risk is required for irreversible blacklisting, so the same threshold need not serve simultaneously as a contribution filter and a permanent-exclusion rule.

Let B_k collect the fixed admission, content, and history factors in $RawScore_k = B_k(1 - r)^p$. For $p > 0$ and $0 \leq r < 1$,

$$\frac{\partial[B_k(1 - r)^p]}{\partial r} = -pB_k(1 - r)^{p-1} \leq 0, \quad (29)$$

$$r = RiskEMA_k^{(t)}.$$

At $r = 1$, the score is zero. This statement concerns the unnormalized score with the other factors fixed; layer selection and normalization determine the update that is eventually applied. SUSPECT and QUARANTINE are recoverable states, allowing later evidence to revise a temporary restriction. Low similarity alone is therefore not a one-round permanent-ban trigger, but a sustained risk signal can still cause exclusion under the explicit persistence rules.

The empirical evaluation must distinguish the state policy from its evidence inputs. A temporal-policy comparison requires identical probes and a matched tuning budget; a probe ablation requires the same state policy. Final-ban outcomes alone do not establish contribution or recovery during training.

5. Experiments

5.1. Reported Setup

The tests involve twenty admitted training clients and eight additional admission-test identities: five Sybils without valid certificates and three free-riders with invalid workload evidence, rejected before training statistics. The main training setting has fourteen benign and six malicious clients; the boundary stress test has ten of each.

The experimental record follows the original manuscript: Flower 1.5.0 and PyTorch train a CIFAR-10-adapted ResNet-18 with a 3×3 , stride-one first convolution, no initial max pooling, and ten output classes. It reports two Intel/AMD TEE-capable CPU platforms with five NVIDIA A10 GPUs, Ubuntu 20.04.6 LTS, and CUDA 12.8. CPU-side protection covers admission and reporting; GPU training is outside that boundary. Each platform has five runs, for ten platform-seed runs in total. Both platforms use the same five seeds, data partitions, and attack schedules, giving five paired experimental conditions and ten executions. Per-platform aggregates are unavailable in the retained record.

Evidence scope. Tables 3–4 retain the original reported values. The author confirms ten platform-seed runs, final-ban semantics, and the stated transition timing and reference-set conventions. The explicit empty-survivor and positive-part calibration rules remain specification changes without matched repeated-run results. A separate local diagnostic trace is reported in Table 5; its different attack configuration, probe access, and simulated attestation prevent pooling it with the main comparison.

TABLE 2. ORIGINAL EXPERIMENTAL CONFIGURATION.

Category	Setting	Value
Task	Model	ResNet-18
Training	Local epochs; lr; momentum; rounds	3; 0.001; 0.9; 30
Data	Shared / exclusive images	5000 / 45000
Admission	w_a, λ, Q	0.7, 1.0, 0.01
Initial state	$T_0, P_0, RiskEMA_0$	1, 0.25, 0
Content	$\alpha_1, \alpha_2, \beta_{fusion}$	0.6, 0.4, 2.0
Memory	β_h, β_r	0.9, 0.85
States	$(\tau_h, \tau_s, \tau_q, \tau_b)$	(0.26, 0.64, 0.80, 0.90)
Persistence	Review; quarantine; blacklist; recovery	2; 3; 4; 2 submissions
Score	$(\alpha, \beta, \gamma, p)$	(3.0, 1.0, 0.5, 1.0)
Layers	$\mu_{base}, \lambda_s, \tau_p$	0.4, 1.2, 3.0
Sensitivity	(w_p, w_u, w_s)	(0.3, 0.3, 0.4)

The 50,000 CIFAR-10 training images are divided into a 5,000-image shared pool replicated at every client and 45,000 exclusive images. The exclusive partition is IID for clients 0–9, Dirichlet $\alpha = 1.0$ for clients 10–14, and $\alpha = 0.1$ for clients 15–19. For a client with the average exclusive size of 2,250, shared images account for $5000/(5000 + 2250) = 68.97\%$ of its training set. This is a calculation from the stated split, not a measured mean of client-specific ratios. The ratios depend on the actual exclusive sizes. Dirichlet concentration therefore describes only the exclusive portion, not the full local distribution.

Writing $\mathcal{D}_k = \mathcal{D}_{shared} \cup \mathcal{D}_{k,exclusive}$ gives $\sum_k |\mathcal{D}_k| = 145,000$ training records under replication. If p_k uses the full local training size as defined in Section 3, shared images are counted at each client; this differs from the objective over 50,000 unique images. The run configuration must establish which counts generated the retained results. The server additionally has a balanced 500-image proxy from CIFAR-10 test data, with 50 images per class. It is separate from client training and supplied to reference-dependent baselines. Proxy, calibration, and final evaluation indices must be checked for overlap; the available record does not establish that split. FedAvg, Krum, and Trimmed Mean do not use the proxy.

C0 performs untargeted label flipping with $\gamma = 0.5$. C1 poisons 20% of local images with a lower-right 3×3 trigger targeting class 0. C2 applies clean-label poisoning to 20% with the same target. C3 introduces a semantic attribute bias in 20% targeting class 0. C4 flips update signs and C5 scales the update by 100. All six malicious identities fall in the IID group; the setting does not establish robustness when malicious identities occupy the Dirichlet groups. The precise

TABLE 3. REPORTED MAIN COMPARISON. ACCURACY AND MACRO TARGETED ASR ARE PERCENTAGES (MEAN $\pm$ SD OVER TEN PLATFORM-SEED RUNS).

Method / setting	Mechanism	Accuracy	Macro target ASR	Final-ban FPR
FedAvg [2]	No defense	86.15 $\pm$ 1.20	92.34 $\pm$ 5.10	N/A
Krum [3]	Geometric selection	64.20 $\pm$ 2.80	12.15 $\pm$ 0.90	N/A
Trimmed Mean [4]	Coordinate trimming	71.35 $\pm$ 2.40	38.60 $\pm$ 4.20	N/A
FLTrust [5]	Clean reference	81.25 $\pm$ 1.50	15.42 $\pm$ 1.10	N/A
Trust Flow	Dual states + layer gates	92.31 $\pm$ 0.09	10.21 $\pm$ 0.05	0.00
Trust Flow: 50% malicious	Boundary stress, 10/20 malicious	91.81 $\pm$ 0.10	10.46 $\pm$ 0.06	0.00

Final-ban FPR is the fraction of benign clients permanently blacklisted at the end of a run. It is undefined for stateless aggregation baselines (N/A). The original pooling script and per-platform values are unavailable.

TABLE 4. REPORTED STATE OUTCOMES, SEPARATED BY ENDPOINT (%). N/R: NOT REPORTED.

Variant	Final-ban FPR	Final-ban TPR	Review FPR	Review TPR
Single-stream EMA, aggressive	18.25 [†]	95.12 [†]	N/R	N/R
Single-stream EMA, conservative	2.10 [†]	48.33 [†]	N/R	N/R
HistPerf-only	0.15 [†]	21.05 [†]	N/R	N/R
Trust Flow	0.00	100.00	14.79	88.33

[†] Original reported values, with final-ban semantics confirmed by the author; exact numerators, denominators, and pooling remain unresolved. The marked entries cannot be reconstructed from equally weighted final-state counts over ten runs of 14 benign and 6 malicious clients. They are not used to infer a matched-count advantage over the other policies.

C2 perturbation construction/budget and C3 attribute remain required configuration details for this six-attack experiment.

For targeted attack $a \in \{\text{trigger}, \text{clean}, \text{semantic}\}$, let $\mathcal{B}_a$ be the attack-specific test subset and y_a^* its target:

$$ASR_a = \frac{|\{x \in \mathcal{B}_a : \arg \max f(x) = y_a^*\}|}{|\mathcal{B}_a|}, \quad (30)$$

$$ASR_{target} = \frac{1}{3} \sum_a ASR_a. \quad (31)$$

The per-attack test counts and treatment of originally target-class samples are not specified in the available record. The macro ASR is therefore retained as reported, without reconstructing its three constituent values. Untargeted C0, C4, and C5 are assessed using accuracy, loss, and influence, not this targeted ASR definition.

5.2. Reported Main Results

Table 3 preserves the reported means and standard deviations. The original comparison uses the same training budget, client partition, attack proportion, and reference access where applicable. This revision does not add unverified baseline results or assert statistical significance. A contemporary backdoor defense and a temporal detection or reputation method remain necessary matched comparisons. The 50% malicious setting is outside the assumed fraction below 50% and provides no theoretical Byzantine guarantee.

5.3. Separate Implementation Trace

A preserved local trace contains one 30-round run with four attacking clients (C0–C3) and sixteen benign clients. C0 uses 50% label flipping and C1 uses 20% trigger poisoning; C2 and C3 are configured at 50%. In this snapshot,

C2 adds a 3×3 upper-left patch only to selected target-class training images without relabeling. C3 adds Gaussian noise of standard deviation 0.1 to selected normalized inputs and clamps them to $[-1, 1]$; it does not implement the semantic attribute attack specified for the main experiment. Known-trigger server probes are enabled, and attestation and runtime telemetry are simulated. Consequently, this trace is an implementation diagnostic rather than a replication of the main threat configuration.

TABLE 5. SEPARATE LOCAL DIAGNOSTIC TRACE: ONE RUN, FOUR ATTACKERS.

Endpoint	Observed value
Final clean accuracy	92.40%
Lower-right trigger ASR	10.21%
Upper-left trigger ASR	10.22%
Final benign / malicious bans	0/16; 4/4
Benign client-rounds with at least one selected layer	466/480

All twenty client logs agree with the server’s clean accuracy and lower-right ASR history. Both targeted evaluations use all 10,000 test images, including original target-class images; no semantic targeted ASR is recorded. The current data-selection code places all 500 proxy images inside this evaluation set. These values therefore do not establish held-out, non-target-only attack success or a three-attack macro average. The logged final percentages imply 1,021 and 1,022 target predictions, respectively, under these denominators; per-example predictions are not retained.

The layer audit records selected-array counts for each processed client. Sixteen benign clients yield 480 client-round opportunities, of which 466 select at least one array. Selection does not establish a nonzero applied update, be-

cause numerical weights and layer updates are not logged. C3 and C2 first enter aggregation isolation in rounds 9 and 12; the first blacklist interceptions for C0/C1, C3, and C2 occur in rounds 8, 16, and 24. This trace contains no benign aggregation-quarantine event from which to estimate benign post-quarantine recovery time.

5.4. Final Bans, Review, and Contribution

A final-ban rate uses a client-level denominator. A review rate uses eligible client-round events. Neither establishes that a client contributes nonzero aggregation weight. Table 4 keeps these outcomes separate. The first three final-ban pairs are retained under the author’s interpretation and the original manuscript label; their underlying event counts and averaging rule remain to be verified. Missing review results are marked N/R and are not inferred from final outcomes.

The reported Trust Flow final retention is 100%, consistent with zero final benign bans, while its reported review FPR is 14.79%. These values do not imply zero review false alarms or 100% per-round detection. Recovery should be assessed by the fraction of eligible benign submissions receiving positive weight, quarantine duration, and rounds until the first positive contribution after quarantine. Malicious cumulative aggregation weight before isolation is a separate exposure measure. The local trace supplies layer-selection counts, but neither final retention nor those counts supplies applied weights or benign quarantine-recovery times.

The retained single-stream values describe a threshold tradeoff, but the records do not establish equal evidence inputs and tuning budgets across variants. A same-evidence comparison should include history-only, risk-only, single-stream, and dual-stream policies, evaluated at comparable benign restriction levels. The current values cannot establish general superiority of two stored states over a tuned single-stream policy.

5.5. Overhead and Remaining Validation

TABLE 6. ORIGINAL REPORTED COMPONENT COSTS; PLATFORM BREAKDOWN UNAVAILABLE.

Component and scope	Reported value
FedAvg server aggregation	2.10 s
Trust Flow server aggregation + audit	4.85 s
Additional TrustReport per client submission	About 15 KB
Auxiliary security path per admitted-client round	About 11 ms
Quote, signing, monitoring per admitted-client round	About 12–15 ms

Table 6 retains the original component timings. Platform assignment, measurement counts, variability, and possible overlap between the 11 ms and 12–15 ms paths are not established; these values are not summed. The server timing

excludes communication and vendor-specific attestation latency. CPU-side TEE capability is not evidence of GPU protection, nor does one timing set characterize both platforms’ end-to-end deployment costs. The separate local Flower trace reports 3,878.53 s for its complete 30-round run, or 129.28 s per round when amortized. This single run-level timer includes training and evaluation within the Flower execution; it is not a measurement of isolated aggregation, attestation, or either hardware platform.

The current shared pool must be varied independently of the server proxy to assess dependence on common training data. Experiments should compare no shared pool, a smaller pool, and the reported pool, with realized class histograms and malicious identities distributed across the data groups. A Dirichlet sweep on exclusive data alone cannot establish these properties. The loss and activation scales also require calibration records and channel trigger rates for benign IID, benign heterogeneous, and malicious updates.

Validation of the revised gate requires per-layer sensitivity, threshold, score distribution, survivor count, skipped updates, and actual applied weights. A client trajectory must link evidence, state, and nonzero model contribution, rather than showing risk or history alone. The local audit provides client-level layer counts and probe thresholds, but does not retain numerical per-layer weights or complete counterfactual probe results. Missing attack-specific or recovery curves are not reconstructed from plotted points.

6. Discussion and Conclusion

Trust Flow connects execution admission, current content, temporal evidence, and layer-wise influence through separate historical-utility and risk states. The revised specification makes recoverable restrictions, permanent exclusion, and aggregation eligibility distinct. It also avoids bypassing a failed layer gate when no update survives. Sustained geometric disagreement can still drive exclusion, and benign recovery requires subsequent favorable evidence.

The original record reports 92.31% accuracy, 10.21% macro target ASR, and zero final benign bans for 30% mixed attacks. Its scope is CIFAR-10 with a large shared pool, a server proxy, and malicious identities in the IID group. The separate trace makes layer selection and two targeted evaluation denominators inspectable without expanding the scope of the main comparison. Matched validation of the explicit empty-survivor and calibration rules, run-level statistics, actual contribution and recovery, shared-pool sensitivity, and platform-specific TEE instrumentation remain open evaluation requirements.

References

- [1] P. Kairouz, H. B. McMahan, B. Avent *et al.*, “Advances and open problems in federated learning,” *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.
- [2] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agueray Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the 20th International*

- Conference on Artificial Intelligence and Statistics*, vol. 54, 2017, pp. 1273–1282. [Online]. Available: <https://proceedings.mlr.press/v54/mcmahan17a.html>
- [3] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, “Machine learning with adversaries: Byzantine tolerant gradient descent,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: <https://papers.nips.cc/paper/6617-machine-learning-with-adversaries-byzantine-tolerant-gradient-descent>
 - [4] D. Yin, Y. Chen, K. Ramchandran, and P. Bartlett, “Byzantine-robust distributed learning: Towards optimal statistical rates,” in *Proceedings of the 35th International Conference on Machine Learning*, vol. 80, 2018, pp. 5650–5659. [Online]. Available: <https://proceedings.mlr.press/v80/yin18a.html>
 - [5] X. Cao, M. Fang, J. Liu, and N. Z. Gong, “FLTrust: Byzantine-robust federated learning via trust bootstrapping,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2021.
 - [6] C. Fung, C. J. M. Yoon, and I. Beschastnikh, “The limitations of federated learning in Sybil settings,” in *Proceedings of the 23rd International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*, 2020, pp. 301–316. [Online]. Available: <https://www.usenix.org/conference/raid2020/presentation/fung>
 - [7] Z. Zhang, X. Cao, J. Jia, and N. Z. Gong, “FLDetector: Defending federated learning against model poisoning attacks via detecting malicious clients,” in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 2545–2555.
 - [8] P. Rieger, T. D. Nguyen, M. Miettinen, and A. R. Sadeghi, “Deep-Sight: Mitigating backdoor attacks in federated learning through deep model inspection,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2022.
 - [9] T. D. Nguyen, P. Rieger, H. Chen, H. Yalame, H. Möllering, H. Fereidooni, S. Marchal, M. Miettinen, A. Mirhoseini, S. Zeitouni, F. Koushanfar, A.-R. Sadeghi, and T. Schneider, “FLAME: Taming backdoors in federated learning,” in *Proceedings of the 31st USENIX Security Symposium*, 2022, pp. 1415–1432. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity22/presentation/nguyen>
 - [10] P. Rieger, T. Krauß, M. Miettinen, A. Dmitrienko, and A. R. Sadeghi, “CrowdGuard: Federated backdoor detection in federated learning,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2024.
 - [11] H. Yang, W. Xi, Y. Shen, C. Wu, and J. Zhao, “RoseAgg: Robust defense against targeted collusion attacks in federated learning,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 2951–2966, 2024.
 - [12] Y. Chen, F. Luo, T. Li, T. Xiang, Z. Liu, and J. Li, “A training-integrity privacy-preserving federated learning scheme with trusted execution environment,” *Information Sciences*, vol. 522, pp. 69–79, 2020.
 - [13] T. Xu, K. Zhu, A. Andrzejak, and L. Zhang, “Distributed learning in trusted execution environment: A case study of federated learning in SGX,” in *Proceedings of the 7th IEEE International Conference on Network Intelligence and Digital Content (IC-NIDC)*, 2021, pp. 450–454.
 - [14] S. Queyruet, V. Schiavoni, and P. Felber, “Mitigating adversarial attacks in federated learning with trusted execution environments,” in *Proceedings of the IEEE 43rd International Conference on Distributed Computing Systems (ICDCS)*, 2023, pp. 626–637.
 - [15] Z. Lu, L. Wang, Z. Zhang, M. Huang, J. Wang, and M. Li, “TMT-FL: Enabling trustworthy model training of federated learning with malicious participants,” *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 3, pp. 2723–2740, 2025.
 - [16] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, “How To Backdoor Federated Learning,” in *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, vol. 108, 2020, pp. 2938–2948. [Online]. Available: <https://proceedings.mlr.press/v108/bagdasaryan20a.html>
 - [17] B. Wang, Y. Yao, S. Shan, H. Li, B. Viswanath, H. Zheng, and B. Y. Zhao, “Neural Cleanse: Identifying and mitigating backdoor attacks in neural networks,” in *Proceedings of the IEEE Symposium on Security and Privacy (SP)*, 2019, pp. 707–723.
 - [18] J. Zhang, C. Zhu, X. Sun, C. Ge, B. Chen, W. Susilo, and S. Yu, “FLPurifier: Backdoor defense in federated learning via decoupled contrastive training,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 4752–4766, 2024.
 - [19] Z. Ma, J. Ma, Y. Miao, Y. Li, and R. H. Deng, “ShieldFL: Mitigating model poisoning attacks in privacy-preserving federated learning,” *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 1639–1654, 2022.
 - [20] L. Wang, M. Huang, Z. Zhang, M. Li, J. Wang, and K. Gai, “RaSA: Robust and adaptive secure aggregation for edge-assisted hierarchical federated learning,” *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 4280–4295, 2025.
 - [21] X. Zhang, Z. Chen, G. Chen, X. Feng, Q. Shen, and Z. Wu, “RPPFL: Robust and privacy-preserving federated learning via trusted execution environments,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.